Total No. of Questions : 8] [Total No. of Printed Pages : 3

Paper Code : 21316

F-416

B.C.A. (Fourth Semester) Examination, 2024

(New Course)

Paper No. : BCA-N-401

OPERATING SYSTEM

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) Define.
   (i) Multiprogramming
   (ii) Multitasking
   (iii) Multiprocessing

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   (iv) Time sharing
   (v) Virtual memory

   (b) What are the major activities of an operating system with regard to process management?

2. (a) Define demand paging in memory management with example.
   (b) Define C-scan scheduling in detail.

3. (a) Distinguish between User file directory (UFD) and Master file directory (MFD)
   (b) Explain Virtual memory. How it is implemented?

4. (a) Describe various types of allocation methods with examples.
   (b) What is deadlock? How do you deal with potential deadlock?

5. (a) Describe the various layers of file system.
   (b) Explain various disk scheduling algorithms with illustrations.

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6. (a) Differentiate between multiprogramming with fixed partition and multiprogramming with variable partition.

(b) Explain the deadlock detection algorithm for single and multiple instance type

7. (a) What are the advantages of indexed allocation? How can the index blocks be implemented in the indexed allocation scheme?

(b) Explain various page replacement strategies.

8. (a) Explain the concept of deadlock recovery

(b) Explain the following allocation algorithms

(i) First-Fit

(ii) Best-Fit

(iii) Worst-Fit

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